

# DSA Patterns

Complete LeetCode Question List

## About ME

Hey, I'm Shubham, 7+ years in tech, 100+ interviews on both sides of the table.

**Career path:** Infosys → Pubmatic → Amazon → Coursera (currently WFH)

I've cracked multiple high-paying offers across India and internationally using the exact playbook below.

Connect here - [Linkedin](#) , [Instagram](#),

Connect on - [Whatsapp](#)

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by Codersoni

*Created by Shubham Soni*

Patterns -

<https://whimsical.com/shubham-s-workspace17856/frontend-NkNqHYxFGoH73oz3bRYa6g>

## ARRAY

### Two Pointer

| Problem                       | URL                                                                                                                                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 167. Two Sum II               | <a href="https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/">https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/</a> |
| 15. 3Sum                      | <a href="https://leetcode.com/problems/3sum/">https://leetcode.com/problems/3sum/</a>                                                         |
| 11. Container With Most Water | <a href="https://leetcode.com/problems/container-with-most-water/">https://leetcode.com/problems/container-with-most-water/</a>               |

### Sliding Window

| Problem                                           | URL                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 643. Maximum Average Subarray I                   | <a href="https://leetcode.com/problems/maximum-average-subarray-i/">https://leetcode.com/problems/maximum-average-subarray-i/</a>                                         |
| 3. Longest Substring Without Repeating Characters | <a href="https://leetcode.com/problems/longest-substring-without-repeating-characters/">https://leetcode.com/problems/longest-substring-without-repeating-characters/</a> |
| 76. Minimum Window Substring                      | <a href="https://leetcode.com/problems/minimum-window-substring/">https://leetcode.com/problems/minimum-window-substring/</a>                                             |

### Prefix Sum

| Problem                    | URL                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 560. Subarray Sum Equals K | <a href="https://leetcode.com/problems/subarray-sum-equals-k/">https://leetcode.com/problems/subarray-sum-equals-k/</a> |

|                                   |                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 238. Product of Array Except Self | <a href="https://leetcode.com/problems/product-of-array-except-self/">https://leetcode.com/problems/product-of-array-except-self/</a> |
| 525. Contiguous Array             | <a href="https://leetcode.com/problems/contiguous-array/">https://leetcode.com/problems/contiguous-array/</a>                         |

## Kadane's

| Problem                            | URL                                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 53. Maximum Subarray               | <a href="https://leetcode.com/problems/maximum-subarray/">https://leetcode.com/problems/maximum-subarray/</a>                           |
| 152. Maximum Product Subarray      | <a href="https://leetcode.com/problems/maximum-product-subarray/">https://leetcode.com/problems/maximum-product-subarray/</a>           |
| 918. Maximum Sum Circular Subarray | <a href="https://leetcode.com/problems/maximum-sum-circular-subarray/">https://leetcode.com/problems/maximum-sum-circular-subarray/</a> |

## Binary Search

| Problem                            | URL                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 704. Binary Search                 | <a href="https://leetcode.com/problems/binary-search/">https://leetcode.com/problems/binary-search/</a>                                   |
| 33. Search in Rotated Sorted Array | <a href="https://leetcode.com/problems/search-in-rotated-sorted-array/">https://leetcode.com/problems/search-in-rotated-sorted-array/</a> |
| 410. Split Array Largest Sum       | <a href="https://leetcode.com/problems/split-array-largest-sum/">https://leetcode.com/problems/split-array-largest-sum/</a>               |

# STRING

## Sliding Window

| Problem                                           | URL                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. Longest Substring Without Repeating Characters | <a href="https://leetcode.com/problems/longest-substring-without-repeating-characters/">https://leetcode.com/problems/longest-substring-without-repeating-characters/</a> |
| 76. Minimum Window Substring                      | <a href="https://leetcode.com/problems/minimum-window-substring/">https://leetcode.com/problems/minimum-window-substring/</a>                                             |
| 567. Permutation in String                        | <a href="https://leetcode.com/problems/permutation-in-string/">https://leetcode.com/problems/permutation-in-string/</a>                                                   |

## Two Pointers

| Problem                      | URL                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 125. Valid Palindrome        | <a href="https://leetcode.com/problems/valid-palindrome/">https://leetcode.com/problems/valid-palindrome/</a>                   |
| 151. Reverse Words in String | <a href="https://leetcode.com/problems/reverse-words-in-a-string/">https://leetcode.com/problems/reverse-words-in-a-string/</a> |
| 443. String Compression      | <a href="https://leetcode.com/problems/string-compression/">https://leetcode.com/problems/string-compression/</a>               |

## Pattern Matching

| Problem                            | URL                                                                                                                                                                               |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28. Find Index of First Occurrence | <a href="https://leetcode.com/problems/find-the-index-of-the-first-occurrence-in-a-string/">https://leetcode.com/problems/find-the-index-of-the-first-occurrence-in-a-string/</a> |
| 686. Repeated String Match         | <a href="https://leetcode.com/problems/repeated-string-match/">https://leetcode.com/problems/repeated-string-match/</a>                                                           |
| 459. Repeated Substring Pattern    | <a href="https://leetcode.com/problems/repeated-substring-pattern/">https://leetcode.com/problems/repeated-substring-pattern/</a>                                                 |

# HASH MAP

## Frequency Based

| Problem            | URL                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------|
| 1. Two Sum         | <a href="https://leetcode.com/problems/two-sum/">https://leetcode.com/problems/two-sum/</a>               |
| 49. Group Anagrams | <a href="https://leetcode.com/problems/group-anagrams/">https://leetcode.com/problems/group-anagrams/</a> |

|                              |                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 347. Top K Frequent Elements | <a href="https://leetcode.com/problems/top-k-frequent-elements/">https://leetcode.com/problems/top-k-frequent-elements/</a> |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|

## Lookup / Set Based

| Problem                           | URL                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 128. Longest Consecutive Sequence | <a href="https://leetcode.com/problems/longest-consecutive-sequence/">https://leetcode.com/problems/longest-consecutive-sequence/</a> |
| 217. Contains Duplicate           | <a href="https://leetcode.com/problems/contains-duplicate/">https://leetcode.com/problems/contains-duplicate/</a>                     |
| 380. Insert Delete GetRandom O(1) | <a href="https://leetcode.com/problems/insert-delete-getrandom-o1/">https://leetcode.com/problems/insert-delete-getrandom-o1/</a>     |

## STACK

### Monotonic Stack

| Problem                            | URL                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 739. Daily Temperatures            | <a href="https://leetcode.com/problems/daily-temperatures/">https://leetcode.com/problems/daily-temperatures/</a>                         |
| 84. Largest Rectangle in Histogram | <a href="https://leetcode.com/problems/largest-rectangle-in-histogram/">https://leetcode.com/problems/largest-rectangle-in-histogram/</a> |
| 901. Online Stock Span             | <a href="https://leetcode.com/problems/online-stock-span/">https://leetcode.com/problems/online-stock-span/</a>                           |

### Nearest Element

| Problem                      | URL                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 496. Next Greater Element I  | <a href="https://leetcode.com/problems/next-greater-element-i/">https://leetcode.com/problems/next-greater-element-i/</a>   |
| 503. Next Greater Element II | <a href="https://leetcode.com/problems/next-greater-element-ii/">https://leetcode.com/problems/next-greater-element-ii/</a> |
| 42. Trapping Rain Water      | <a href="https://leetcode.com/problems/trapping-rain-water/">https://leetcode.com/problems/trapping-rain-water/</a>         |

### Expression Handling

| Problem                               | URL                                                                                                                                           |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 20. Valid Parentheses                 | <a href="https://leetcode.com/problems/valid-parentheses/">https://leetcode.com/problems/valid-parentheses/</a>                               |
| 150. Evaluate Reverse Polish Notation | <a href="https://leetcode.com/problems/evaluate-reverse-polish-notation/">https://leetcode.com/problems/evaluate-reverse-polish-notation/</a> |
| 224. Basic Calculator                 | <a href="https://leetcode.com/problems/basic-calculator/">https://leetcode.com/problems/basic-calculator/</a>                                 |

## QUEUE / DEQUE

| Problem                                    | URL                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 933. Number of Recent Calls                | <a href="https://leetcode.com/problems/number-of-recent-calls/">https://leetcode.com/problems/number-of-recent-calls/</a>                               |
| 239. Sliding Window Maximum                | <a href="https://leetcode.com/problems/sliding-window-maximum/">https://leetcode.com/problems/sliding-window-maximum/</a>                               |
| 862. Shortest Subarray with Sum at Least K | <a href="https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/">https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/</a> |

## LINKED LIST

### Fast-Slow Pointer

| Problem                   | URL                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 141. Linked List Cycle    | <a href="https://leetcode.com/problems/linked-list-cycle/">https://leetcode.com/problems/linked-list-cycle/</a>       |
| 142. Linked List Cycle II | <a href="https://leetcode.com/problems/linked-list-cycle-ii/">https://leetcode.com/problems/linked-list-cycle-ii/</a> |

|                                |                                                                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 876. Middle of the Linked List | <a href="https://leetcode.com/problems/middle-of-the-linked-list/">https://leetcode.com/problems/middle-of-the-linked-list/</a> |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|

## Reversal

| Problem                      | URL                                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 206. Reverse Linked List     | <a href="https://leetcode.com/problems/reverse-linked-list/">https://leetcode.com/problems/reverse-linked-list/</a>           |
| 92. Reverse Linked List II   | <a href="https://leetcode.com/problems/reverse-linked-list-ii/">https://leetcode.com/problems/reverse-linked-list-ii/</a>     |
| 25. Reverse Nodes in k-Group | <a href="https://leetcode.com/problems/reverse-nodes-in-k-group/">https://leetcode.com/problems/reverse-nodes-in-k-group/</a> |

## Merge

| Problem                    | URL                                                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 21. Merge Two Sorted Lists | <a href="https://leetcode.com/problems/merge-two-sorted-lists/">https://leetcode.com/problems/merge-two-sorted-lists/</a> |
| 23. Merge K Sorted Lists   | <a href="https://leetcode.com/problems/merge-k-sorted-lists/">https://leetcode.com/problems/merge-k-sorted-lists/</a>     |
| 148. Sort List             | <a href="https://leetcode.com/problems/sort-list/">https://leetcode.com/problems/sort-list/</a>                           |

# TREES

## DFS

| Problem                           | URL                                                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 104. Maximum Depth of Binary Tree | <a href="https://leetcode.com/problems/maximum-depth-of-binary-tree/">https://leetcode.com/problems/maximum-depth-of-binary-tree/</a> |
| 112. Path Sum                     | <a href="https://leetcode.com/problems/path-sum/">https://leetcode.com/problems/path-sum/</a>                                         |
| 543. Diameter of Binary Tree      | <a href="https://leetcode.com/problems/diameter-of-binary-tree/">https://leetcode.com/problems/diameter-of-binary-tree/</a>           |

## BFS

| Problem                                | URL                                                                                                                                                           |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 102. Binary Tree Level Order Traversal | <a href="https://leetcode.com/problems/binary-tree-level-order-traversal/">https://leetcode.com/problems/binary-tree-level-order-traversal/</a>               |
| 103. Binary Tree Zigzag Level Order    | <a href="https://leetcode.com/problems/binary-tree-zigzag-level-order-traversal/">https://leetcode.com/problems/binary-tree-zigzag-level-order-traversal/</a> |
| 199. Binary Tree Right Side View       | <a href="https://leetcode.com/problems/binary-tree-right-side-view/">https://leetcode.com/problems/binary-tree-right-side-view/</a>                           |

## Path Based

| Problem                               | URL                                                                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 124. Binary Tree Maximum Path Sum     | <a href="https://leetcode.com/problems/binary-tree-maximum-path-sum/">https://leetcode.com/problems/binary-tree-maximum-path-sum/</a>       |
| 257. Binary Tree Paths                | <a href="https://leetcode.com/problems/binary-tree-paths/">https://leetcode.com/problems/binary-tree-paths/</a>                             |
| 1448. Count Good Nodes in Binary Tree | <a href="https://leetcode.com/problems/count-good-nodes-in-binary-tree/">https://leetcode.com/problems/count-good-nodes-in-binary-tree/</a> |

## BST

| Problem                            | URL                                                                                                                                                                       |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 98. Validate Binary Search Tree    | <a href="https://leetcode.com/problems/validate-binary-search-tree/">https://leetcode.com/problems/validate-binary-search-tree/</a>                                       |
| 230. Kth Smallest Element in BST   | <a href="https://leetcode.com/problems/kth-smallest-element-in-a-bst/">https://leetcode.com/problems/kth-smallest-element-in-a-bst/</a>                                   |
| 235. Lowest Common Ancestor of BST | <a href="https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree/">https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree/</a> |

# RECURSION / BACKTRACKING

## Subsets

| Problem                      | URL                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 78. Subsets                  | <a href="https://leetcode.com/problems/subsets/">https://leetcode.com/problems/subsets/</a>                                 |
| 90. Subsets II               | <a href="https://leetcode.com/problems/subsets-ii/">https://leetcode.com/problems/subsets-ii/</a>                           |
| 131. Palindrome Partitioning | <a href="https://leetcode.com/problems/palindrome-partitioning/">https://leetcode.com/problems/palindrome-partitioning/</a> |

## Permutations

| Problem                  | URL                                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 46. Permutations         | <a href="https://leetcode.com/problems/permutations/">https://leetcode.com/problems/permutations/</a>                 |
| 47. Permutations II      | <a href="https://leetcode.com/problems/permutations-ii/">https://leetcode.com/problems/permutations-ii/</a>           |
| 60. Permutation Sequence | <a href="https://leetcode.com/problems/permutation-sequence/">https://leetcode.com/problems/permutation-sequence/</a> |

## Combinations

| Problem                | URL                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| 77. Combinations       | <a href="https://leetcode.com/problems/combinations/">https://leetcode.com/problems/combinations/</a>             |
| 39. Combination Sum    | <a href="https://leetcode.com/problems/combination-sum/">https://leetcode.com/problems/combination-sum/</a>       |
| 40. Combination Sum II | <a href="https://leetcode.com/problems/combination-sum-ii/">https://leetcode.com/problems/combination-sum-ii/</a> |

## Word Search

| Problem                 | URL                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| 79. Word Search         | <a href="https://leetcode.com/problems/word-search/">https://leetcode.com/problems/word-search/</a>               |
| 212. Word Search II     | <a href="https://leetcode.com/problems/word-search-ii/">https://leetcode.com/problems/word-search-ii/</a>         |
| 489. Robot Room Cleaner | <a href="https://leetcode.com/problems/robot-room-cleaner/">https://leetcode.com/problems/robot-room-cleaner/</a> |

## Divide & Conquer

| Problem                       | URL                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 148. Sort List                | <a href="https://leetcode.com/problems/sort-list/">https://leetcode.com/problems/sort-list/</a>                                                     |
| 215. Kth Largest Element      | <a href="https://leetcode.com/problems/kth-largest-element-in-an-array/">https://leetcode.com/problems/kth-largest-element-in-an-array/</a>         |
| 315. Count of Smaller Numbers | <a href="https://leetcode.com/problems/count-of-smaller-numbers-after-self/">https://leetcode.com/problems/count-of-smaller-numbers-after-self/</a> |

## HEAP

### Top K

| Problem                         | URL                                                                                                                                         |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 215. Kth Largest Element        | <a href="https://leetcode.com/problems/kth-largest-element-in-an-array/">https://leetcode.com/problems/kth-largest-element-in-an-array/</a> |
| 347. Top K Frequent Elements    | <a href="https://leetcode.com/problems/top-k-frequent-elements/">https://leetcode.com/problems/top-k-frequent-elements/</a>                 |
| 973. K Closest Points to Origin | <a href="https://leetcode.com/problems/k-closest-points-to-origin/">https://leetcode.com/problems/k-closest-points-to-origin/</a>           |

### Greedy + Heap

| Problem                | URL                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------|
| 621. Task Scheduler    | <a href="https://leetcode.com/problems/task-scheduler/">https://leetcode.com/problems/task-scheduler/</a>       |
| 767. Reorganize String | <a href="https://leetcode.com/problems/reorganize-string/">https://leetcode.com/problems/reorganize-string/</a> |
| 253. Meeting Rooms II  | <a href="https://leetcode.com/problems/meeting-rooms-ii/">https://leetcode.com/problems/meeting-rooms-ii/</a>   |

### K-way Merge

| Problem                                    | URL                                                                                                                                                                     |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23. Merge K Sorted Lists                   | <a href="https://leetcode.com/problems/merge-k-sorted-lists/">https://leetcode.com/problems/merge-k-sorted-lists/</a>                                                   |
| 378. Kth Smallest Element in Sorted Matrix | <a href="https://leetcode.com/problems/kth-smallest-element-in-a-sorted-matrix/">https://leetcode.com/problems/kth-smallest-element-in-a-sorted-matrix/</a>             |
| 632. Smallest Range Covering K Lists       | <a href="https://leetcode.com/problems/smallest-range-covering-elements-from-k-lists/">https://leetcode.com/problems/smallest-range-covering-elements-from-k-lists/</a> |

## GRAPHS

### BFS / DFS

| Problem                 | URL                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| 200. Number of Islands  | <a href="https://leetcode.com/problems/number-of-islands/">https://leetcode.com/problems/number-of-islands/</a>   |
| 133. Clone Graph        | <a href="https://leetcode.com/problems/clone-graph/">https://leetcode.com/problems/clone-graph/</a>               |
| 695. Max Area of Island | <a href="https://leetcode.com/problems/max-area-of-island/">https://leetcode.com/problems/max-area-of-island/</a> |

### Cycle Detection

| Problem                        | URL                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 207. Course Schedule           | <a href="https://leetcode.com/problems/course-schedule/">https://leetcode.com/problems/course-schedule/</a>                     |
| 684. Redundant Connection      | <a href="https://leetcode.com/problems/redundant-connection/">https://leetcode.com/problems/redundant-connection/</a>           |
| 802. Find Eventual Safe States | <a href="https://leetcode.com/problems/find-eventual-safe-states/">https://leetcode.com/problems/find-eventual-safe-states/</a> |

### Topological Sort

| Problem                 | URL                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| 207. Course Schedule    | <a href="https://leetcode.com/problems/course-schedule/">https://leetcode.com/problems/course-schedule/</a>       |
| 210. Course Schedule II | <a href="https://leetcode.com/problems/course-schedule-ii/">https://leetcode.com/problems/course-schedule-ii/</a> |
| 269. Alien Dictionary   | <a href="https://leetcode.com/problems/alien-dictionary/">https://leetcode.com/problems/alien-dictionary/</a>     |

### Shortest Path

| Problem                              | URL                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 743. Network Delay Time              | <a href="https://leetcode.com/problems/network-delay-time/">https://leetcode.com/problems/network-delay-time/</a>                                                                                                                   |
| 787. Cheapest Flights Within K Stops | <a href="https://leetcode.com/problems/cheapest-flights-within-k-stops/">https://leetcode.com/problems/cheapest-flights-within-k-stops/</a>                                                                                         |
| 1334. Find the City                  | <a href="https://leetcode.com/problems/find-the-city-with-the-smallest-number-of-neighbors-at-a-threshold-distance/">https://leetcode.com/problems/find-the-city-with-the-smallest-number-of-neighbors-at-a-threshold-distance/</a> |

### Union Find

| Problem                                              | URL                                                                                                                                                                       |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 547. Number of Provinces                             | <a href="https://leetcode.com/problems/number-of-provinces/">https://leetcode.com/problems/number-of-provinces/</a>                                                       |
| 684. Redundant Connection                            | <a href="https://leetcode.com/problems/redundant-connection/">https://leetcode.com/problems/redundant-connection/</a>                                                     |
| 1319. Number of Operations to Make Network Connected | <a href="https://leetcode.com/problems/number-of-operations-to-make-network-connected/">https://leetcode.com/problems/number-of-operations-to-make-network-connected/</a> |

## TRIE

| Problem | URL |
|---------|-----|
|---------|-----|

|                                 |                                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 208. Implement Trie             | <a href="https://leetcode.com/problems/implement-trie-prefix-tree/">https://leetcode.com/problems/implement-trie-prefix-tree/</a>                         |
| 212. Word Search II             | <a href="https://leetcode.com/problems/word-search-ii/">https://leetcode.com/problems/word-search-ii/</a>                                                 |
| 421. Maximum XOR of Two Numbers | <a href="https://leetcode.com/problems/maximum-xor-of-two-numbers-in-an-array/">https://leetcode.com/problems/maximum-xor-of-two-numbers-in-an-array/</a> |

## DYNAMIC PROGRAMMING

### 1D DP

| Problem             | URL                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------|
| 70. Climbing Stairs | <a href="https://leetcode.com/problems/climbing-stairs/">https://leetcode.com/problems/climbing-stairs/</a> |
| 198. House Robber   | <a href="https://leetcode.com/problems/house-robber/">https://leetcode.com/problems/house-robber/</a>       |
| 322. Coin Change    | <a href="https://leetcode.com/problems/coin-change/">https://leetcode.com/problems/coin-change/</a>         |

### 2D / Grid DP

| Problem              | URL                                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------------|
| 62. Unique Paths     | <a href="https://leetcode.com/problems/unique-paths/">https://leetcode.com/problems/unique-paths/</a>         |
| 64. Minimum Path Sum | <a href="https://leetcode.com/problems/minimum-path-sum/">https://leetcode.com/problems/minimum-path-sum/</a> |
| 72. Edit Distance    | <a href="https://leetcode.com/problems/edit-distance/">https://leetcode.com/problems/edit-distance/</a>       |

### Knapsack

| Problem                         | URL                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 416. Partition Equal Subset Sum | <a href="https://leetcode.com/problems/partition-equal-subset-sum/">https://leetcode.com/problems/partition-equal-subset-sum/</a> |
| 494. Target Sum                 | <a href="https://leetcode.com/problems/target-sum/">https://leetcode.com/problems/target-sum/</a>                                 |
| 474. Ones and Zeroes            | <a href="https://leetcode.com/problems/ones-and-zeroes/">https://leetcode.com/problems/ones-and-zeroes/</a>                       |

### Sequence DP

| Problem                              | URL                                                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 300. Longest Increasing Subsequence  | <a href="https://leetcode.com/problems/longest-increasing-subsequence/">https://leetcode.com/problems/longest-increasing-subsequence/</a>   |
| 1143. Longest Common Subsequence     | <a href="https://leetcode.com/problems/longest-common-subsequence/">https://leetcode.com/problems/longest-common-subsequence/</a>           |
| 516. Longest Palindromic Subsequence | <a href="https://leetcode.com/problems/longest-palindromic-subsequence/">https://leetcode.com/problems/longest-palindromic-subsequence/</a> |

### Interval DP

| Problem                              | URL                                                                                                                                                       |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 312. Burst Balloons                  | <a href="https://leetcode.com/problems/burst-balloons/">https://leetcode.com/problems/burst-balloons/</a>                                                 |
| 1039. Minimum Score Triangulation    | <a href="https://leetcode.com/problems/minimum-score-triangulation-of-polygon/">https://leetcode.com/problems/minimum-score-triangulation-of-polygon/</a> |
| 375. Guess Number Higher or Lower II | <a href="https://leetcode.com/problems/guess-number-higher-or-lower-ii/">https://leetcode.com/problems/guess-number-higher-or-lower-ii/</a>               |

### Bitmask DP

| Problem                                      | URL                                                                                                                                                               |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 847. Shortest Path Visiting All Nodes        | <a href="https://leetcode.com/problems/shortest-path-visiting-all-nodes/">https://leetcode.com/problems/shortest-path-visiting-all-nodes/</a>                     |
| 526. Beautiful Arrangement                   | <a href="https://leetcode.com/problems/beautiful-arrangement/">https://leetcode.com/problems/beautiful-arrangement/</a>                                           |
| 1986. Min Cost to Reach City With K Highways | <a href="https://leetcode.com/problems/minimum-cost-to-reach-city-with-no-traffic/">https://leetcode.com/problems/minimum-cost-to-reach-city-with-no-traffic/</a> |

## GREEDY

### Interval

| Problem                        | URL                                                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 56. Merge Intervals            | <a href="https://leetcode.com/problems/merge-intervals/">https://leetcode.com/problems/merge-intervals/</a>                                                       |
| 435. Non-overlapping Intervals | <a href="https://leetcode.com/problems/non-overlapping-intervals/">https://leetcode.com/problems/non-overlapping-intervals/</a>                                   |
| 452. Minimum Number of Arrows  | <a href="https://leetcode.com/problems/minimum-number-of-arrows-to-burst-balloons/">https://leetcode.com/problems/minimum-number-of-arrows-to-burst-balloons/</a> |

### Jump Game

| Problem             | URL                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------|
| 55. Jump Game       | <a href="https://leetcode.com/problems/jump-game/">https://leetcode.com/problems/jump-game/</a>         |
| 45. Jump Game II    | <a href="https://leetcode.com/problems/jump-game-ii/">https://leetcode.com/problems/jump-game-ii/</a>   |
| 1306. Jump Game III | <a href="https://leetcode.com/problems/jump-game-iii/">https://leetcode.com/problems/jump-game-iii/</a> |

### Scheduling

| Problem                        | URL                                                                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 621. Task Scheduler            | <a href="https://leetcode.com/problems/task-scheduler/">https://leetcode.com/problems/task-scheduler/</a>                                                               |
| 1353. Maximum Number of Events | <a href="https://leetcode.com/problems/maximum-number-of-events-that-can-be-attended/">https://leetcode.com/problems/maximum-number-of-events-that-can-be-attended/</a> |
| 630. Course Schedule III       | <a href="https://leetcode.com/problems/course-schedule-iii/">https://leetcode.com/problems/course-schedule-iii/</a>                                                     |

## BIT MANIPULATION

| Problem                  | URL                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|
| 136. Single Number       | <a href="https://leetcode.com/problems/single-number/">https://leetcode.com/problems/single-number/</a>             |
| 191. Number of 1 Bits    | <a href="https://leetcode.com/problems/number-of-1-bits/">https://leetcode.com/problems/number-of-1-bits/</a>       |
| 338. Counting Bits       | <a href="https://leetcode.com/problems/counting-bits/">https://leetcode.com/problems/counting-bits/</a>             |
| 268. Missing Number      | <a href="https://leetcode.com/problems/missing-number/">https://leetcode.com/problems/missing-number/</a>           |
| 371. Sum of Two Integers | <a href="https://leetcode.com/problems/sum-of-two-integers/">https://leetcode.com/problems/sum-of-two-integers/</a> |

## RANGE STRUCTURES

### Segment Tree

| Problem                       | URL                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 307. Range Sum Query Mutable  | <a href="https://leetcode.com/problems/range-sum-query-mutable/">https://leetcode.com/problems/range-sum-query-mutable/</a>                         |
| 315. Count of Smaller Numbers | <a href="https://leetcode.com/problems/count-of-smaller-numbers-after-self/">https://leetcode.com/problems/count-of-smaller-numbers-after-self/</a> |
| 493. Reverse Pairs            | <a href="https://leetcode.com/problems/reverse-pairs/">https://leetcode.com/problems/reverse-pairs/</a>                                             |

### Fenwick Tree

| Problem                      | URL                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 307. Range Sum Query Mutable | <a href="https://leetcode.com/problems/range-sum-query-mutable/">https://leetcode.com/problems/range-sum-query-mutable/</a> |



|                                 |                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 308. Range Sum Query 2D Mutable | <a href="https://leetcode.com/problems/range-sum-query-2d-mutable/">https://leetcode.com/problems/range-sum-query-2d-mutable/</a> |
| 327. Count of Range Sum         | <a href="https://leetcode.com/problems/count-of-range-sum/">https://leetcode.com/problems/count-of-range-sum/</a>                 |